

Unit – 5 : Session – 6 : SLO – 2 | Symbolic Limits and Series**I. Questions & Answers**

1. $\lim_{x \rightarrow \infty} 1/x = 0$ — as x grows without bound, $1/x$ shrinks to 0. In SymPy: `limit(1/x, x, oo) → 0`.
2. Sum of $1 + 1/2 + 1/4 + \dots$ — a geometric series with first term $a = 1$ and ratio $r = 1/2$; $\text{sum} = a/(1 - r) = 1/(1 - 1/2) = 2$.
3. $\lim_{x \rightarrow 0} \sin(x)/x = 1$ — the classic standard limit. In SymPy: `limit(sin(x)/x, x, 0) → 1`.

II. Code

1. `from sympy import symbols, limit, sin, oo, summation`
2. `from sympy.abc import x, n`
3. `print(limit(1/x, x, oo)) # 0`
4. `print(summation((1/2)**n, (n, 0, oo))) # 2`
5. `print(limit(sin(x)/x, x, 0)) # 1`